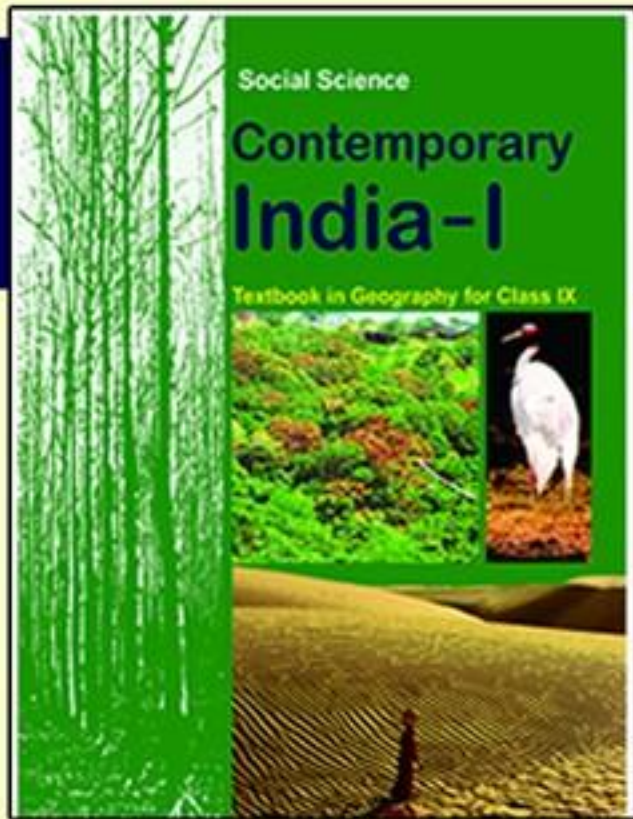




Learning **Space**



NCERT BASICS



GEOGRAPHY

- SHAPE OF THE EARTH
- ROTATION AND REVOLUTION OF THE EARTH

MODULE - 2





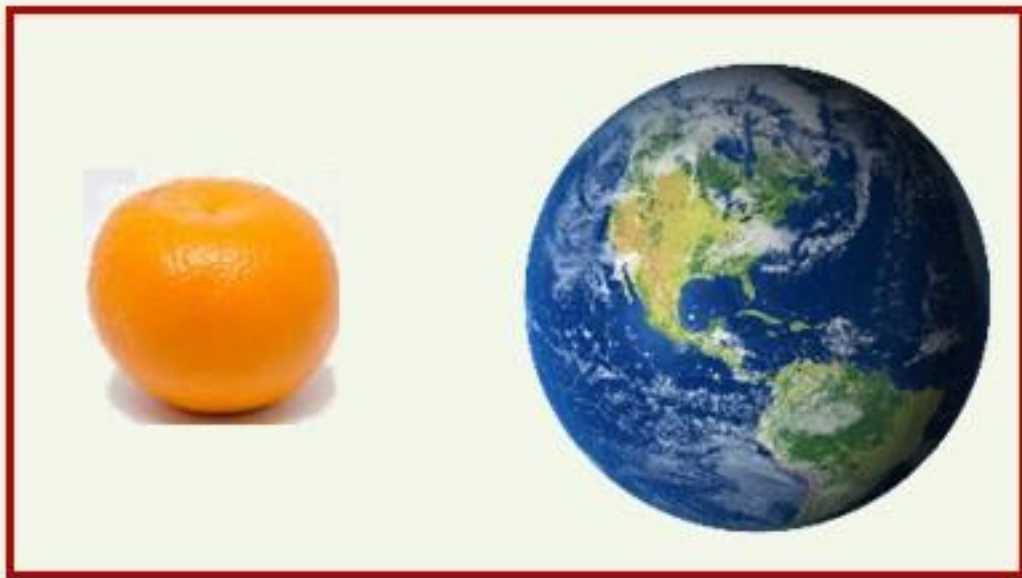
DO YOU KNOW THE EXACT SHAPE OF THE EARTH?

IS THE SHAPE OF EARTH A SPHERE?

Earth is not a perfect sphere, the shape of the earth is called **Geoid**.

WHAT IS GEOID?

- Geoid is the earth like shape.
- It is slightly flattened at the north and south poles and bulges at the equator.
- We calculate precise surface locations considering this shape.





4 IMPORTANT ASPECTS

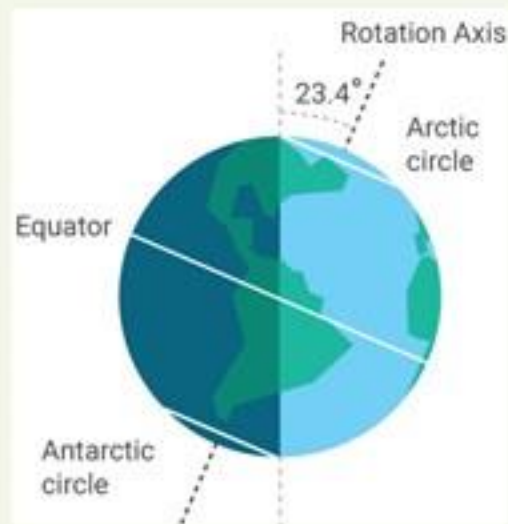
1. TILT TOWARDS ONE SIDE

WHAT DO YOU UNDERSTAND BY OBLIQUITY?

Axial tilt or Obliquity is the angle between the planet's **rotational axis** and its **orbital axis**.

WHAT CAUSED THE EARTH TO TILT?

When an object, the size of Mars crashed into newly formed planet Earth around 4.5 billion years ago, it knocked our planet and left it tilted to an angle.

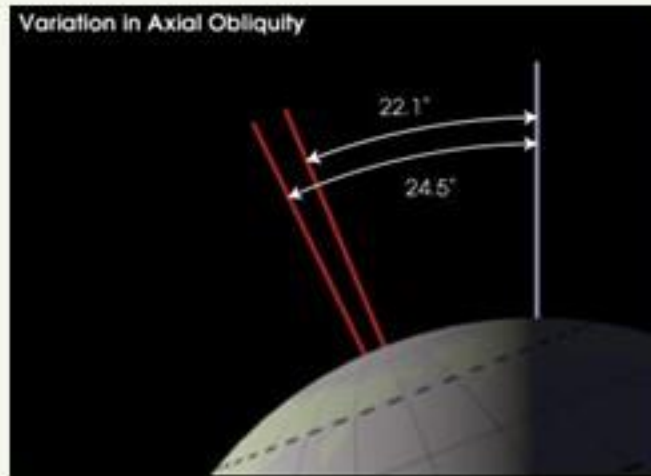


4 IMPORTANT ASPECTS

1. TILT TOWARDS ONE SIDE

IS THE TILT CONSTANT?

- To be specific it's 23.4° tilted at present.
- Earth's axial tilt oscillates between 22.1° and 24.5° .



WHY DOES THE AXIAL TILT ALTER?

- ✓ Earth's axis also wobbles around itself.
- ✓ Also, the gravitational pull of the Moon, Sun and other planets prevents it from shifting by more than a degree.

IMPLICATION OF THE TILT

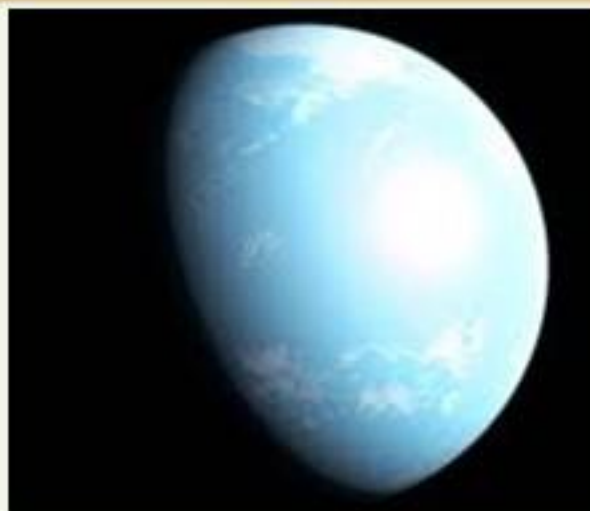
The solar energy reaching different parts of our planet is not constant.

4 IMPORTANT ASPECTS

2. ROTATION OF THE EARTH

SPECIFICS OF ROTATION

- Rotation can be defined as the spinning of the Earth on its own axis.
- Earth rotates from west to east around an imaginary line called axis.
- Earth's rotation is completed in approximately 24 hrs.



REPERCUSSIONS OF ROTATION

- Due to the outward force caused by the rotation of the Earth, it transformed to geoid shape.
- Rotation causes day and nights.

DID YOU KNOW?

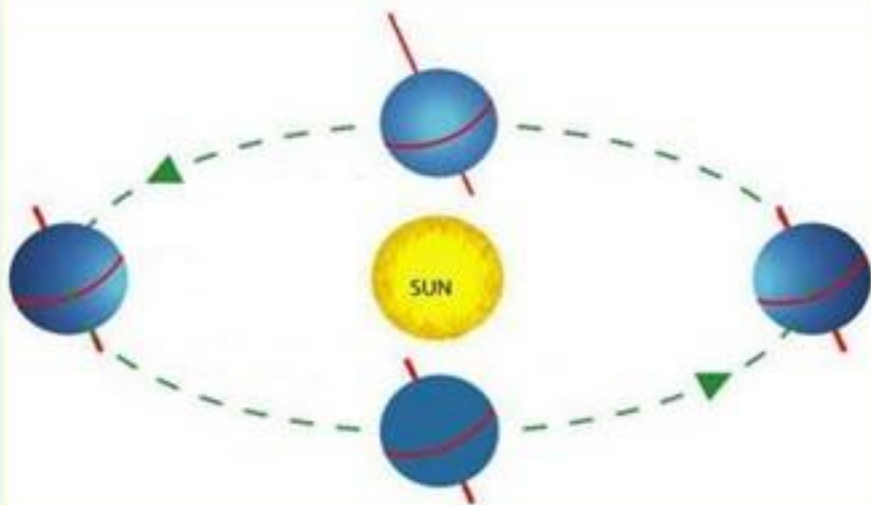
The circle that divides the day and night on the globe is the *circle of illumination*.



4 IMPORTANT ASPECTS

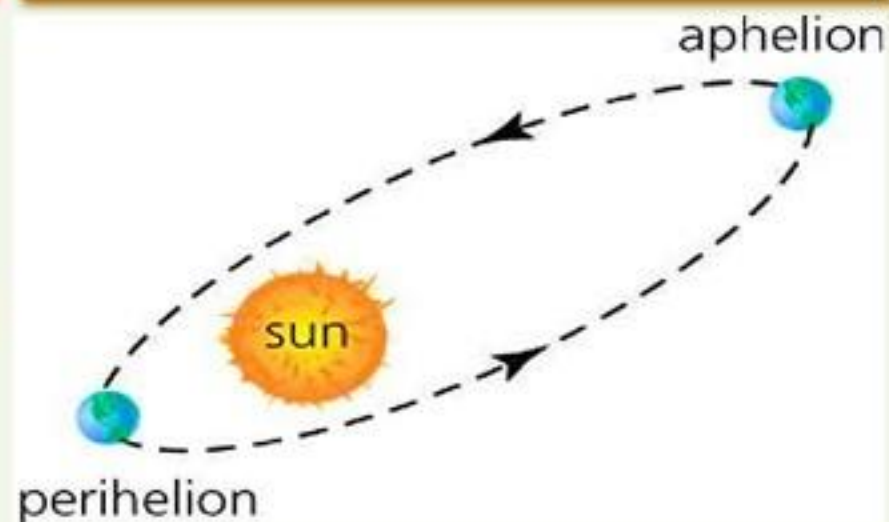
3. REVOLUTION OF EARTH

- Revolution can be defined as movement of Earth around the Sun.
- Earth's revolution takes around $365\frac{1}{4}$ days.



4. ELLIPTICAL PATH

- The path of revolution is not circular.
- Earth revolves around the Sun in an elliptical orbit, with the Sun as its focus.



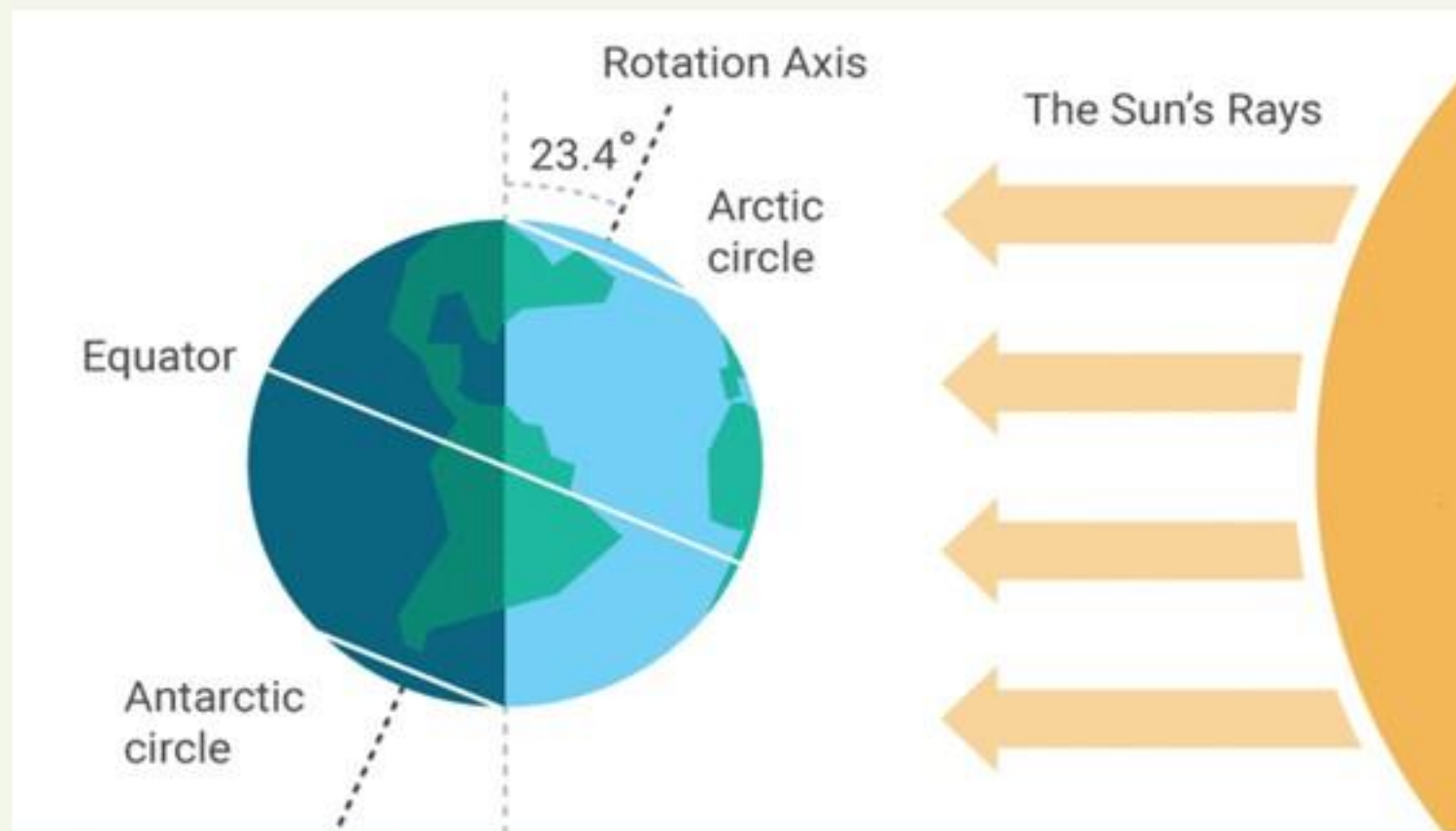


IMPLICATIONS OF THE IMPORTANT ASPECTS

**1. DAYS AND NIGHTS WITH
VARYING DURATION**



**AXIAL TILT +
ROTATION**

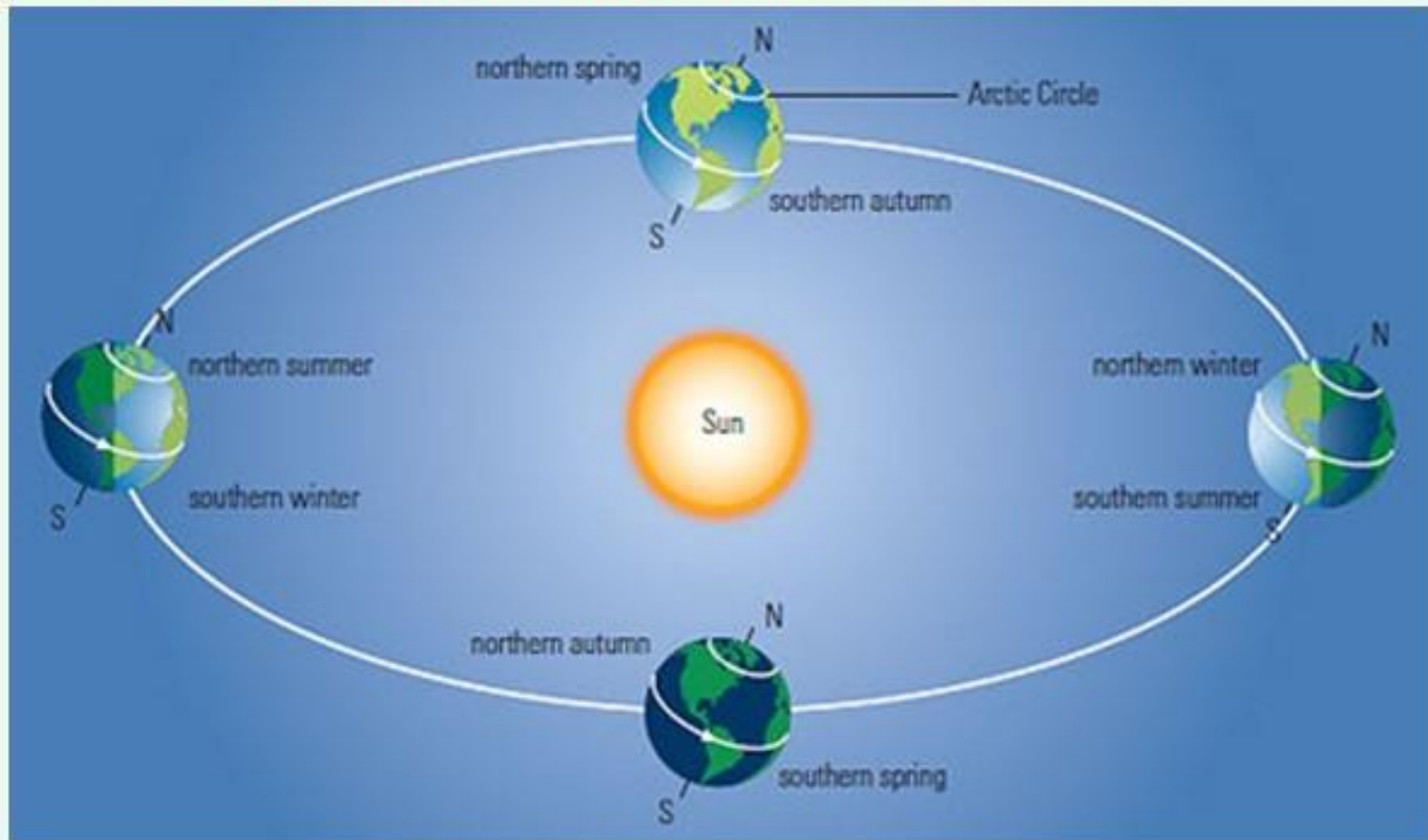


2. DIFFERENT SEASONS

3. SEASONS ARE OPPOSITE IN NORTHERN AND SOUTHERN HEMISPHERE.

AXIAL TILT +
REVOLUTION

Learning Space



Tilt of earth's
axis 23.4°

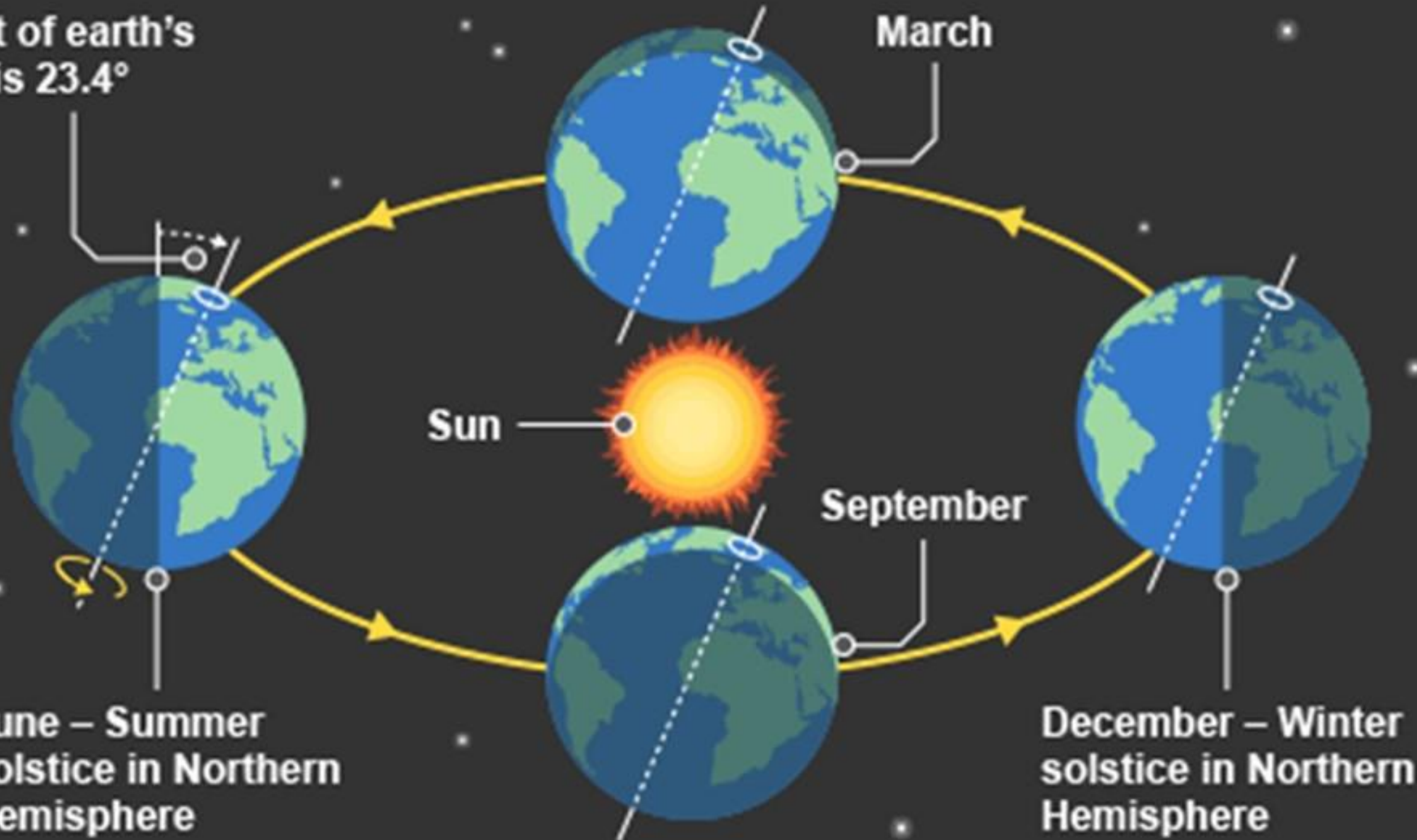
March

Sun

September

June – Summer
solstice in Northern
Hemisphere

December – Winter
solstice in Northern
Hemisphere





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SINCERE THANKS
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